

STNR-Det: Synergistic Temporal Nuance Reconstruction for Remote Sensing Change Detection

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Abstract—Remote sensing change detection (RSCD) identifies surface changes by analyzing bi-temporal images. Existing methods tend to fuse bi-temporal images through simple concatenation or differencing, which provide limited feature integration and poor noise suppression. It finally results in blurred recognition within the transition region between changed and unchanged areas. Considering the above limitations, we propose a Synergistic Temporal Nuance Reconstruction Detector (STNR-Det) for better synergistic bi-temporal feature fusion and fine-grained change delineation, which incorporates two key modules. Firstly, the Contrastive Synergistic Interaction (CSI) Block employs contrast-guided residual modulation to integrate bi-temporal embeddings into a common latent space, which strengthens the capability of difference modeling and synergistic integration. Secondly, the Progressive Nuance Reconstitution (PNR) Decoder is proposed to progressively aggregate features at different granularities, thereby enhancing the representation of transition regions and fine-grained changes. Experiments show that STNR-Det outperforms state-of-the-art methods in accuracy across multiple benchmarks. Ablation studies verify that the CSI block and the PNR decoder are essential for integrating multi-granularity features and suppressing highly similar noise patterns.

Index Terms—Remote sensing, visual place recognition, computer vision, image segmentation, feature extraction.

I. INTRODUCTION

REMOTE sensing change detection (RSCD) aims to identify and quantify changes in surface cover in the spatial and temporal dimensions by comparing and analyzing remotely sensed images of the same region with different time phases [1]–[3]. The technique plays a key role in areas such as environmental evolution monitoring, resource dynamics assessment, urban extension studies and disaster response.

Traditional RSCD methods include pixel-level techniques, such as change vector analysis (CVA) and principal component analysis (PCA) [4], and statistical or feature-based methods such as Kullback-Leibler (KL) divergence [5], which rely on pixel-level or region-level comparisons. However, these approaches struggle with spatial context and noise, making them less robust in complex scenarios. With the advent of deep learning, RSCD methods have significantly improved.

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Early CNN-based methods, such as the FC-EF [6] and SNUNet [7] Siamese models, employ early-fusion strategies and multi-scale processing for bi-temporal feature extraction. Despite improvements in accuracy, CNNs' local receptive fields limit global context modeling and hinder effective interaction and alignment between multi-scale and cross-temporal features, leading to noisy or redundant feature representations. Recent transformer-based methods, like BIT [8] and ChangeFormer [9], leverage self-attention to model long-range dependencies and global semantic relationships, enhancing robustness to varying conditions. However, they still face challenges in preserving local details while maintaining global context, especially for accurate boundary delineation in complex backgrounds. To address these issues, state-space models (SSMs), such as Mamba [10], have shown promise in handling long sequences efficiently, but they still struggle to couple fine-grained local textures with global context [11], [12]. In addition, many aggregation pipelines still rely on simple operations like addition or concatenation, which fail to properly align bi-temporal features, limiting the exchange of complementary cues. As a result, existing methods still suffer from two key limitations: insufficient cross-temporal and multi-scale feature interaction and alignment, and inadequate detail reconstruction in decoding, which often leads to blurred boundaries, incomplete change regions, and pseudo changes.

Motivated by the above limitations, we propose a Synergistic Temporal Nuance Reconstruction Detector (STNR-Det) to establish a clearer link between cross-temporal feature interaction and decoder-side detail reconstruction. To address insufficient temporal interaction, the Contrastive Synergistic Interaction (CSI) Block introduces contrast-guided residual modulation to align bi-temporal and multi-scale features in a shared latent space, thereby enhancing true-change discrimination and suppressing pseudo changes. To alleviate boundary blurring and detail loss, the Progressive Nuance Reconstitution (PNR) Decoder progressively restores transition-region cues and high-frequency details for more complete change regions and sharper contours. The main contributions are summarized as follows:

- 1) A Contrastive Synergistic Interaction (CSI) Block is proposed to enhance the interaction of different-grained, bi-temporal features. It uses contrast-guided feature modulation to align them in a shared latent space and correct spatial/scale inconsistencies, which enables feature-level discrimination of true cross-time changes.
- 2) A Progressive Nuance Reconstitution (PNR) Decoder is introduced to refine decoder-side features through progressive detail reconstitution and transition-region-aware

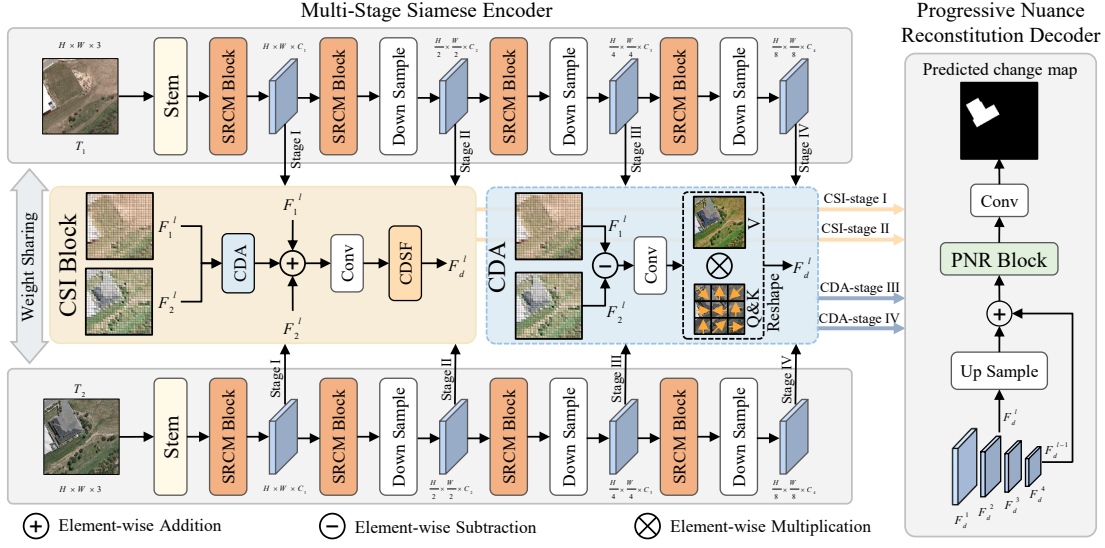


Fig. 1. Overall architecture of the proposed STNR-Det. In the Multi-Stage Siamese Encoder T_1 and T_2 represent bi-temporal images, H, W and C represent the height, width and number of channels of the image, respectively. F_1^l and F_2^l denote the stage- l features extracted from T_1 and T_2 by the Multi-Stage Siamese Encoder, respectively. F_d^l denotes the change-aware feature at stage l generated by the CSI block. In the Progressive Nuance Reconstitution Decoder, F_d^l and F_d^{l-1} represent the features at the current and previous levels.

guidance. It aggregates multi-level features stage by stage and preserves high-frequency transition-region cues. This alleviates boundary information loss and incomplete change contours.

- 3) We construct the Synergistic Temporal Nuance Reconstruction Detector (STNR-Det), which integrates the CSI Block and PNR Decoder into a unified architecture. Qualitative and quantitative studies on three benchmark datasets show that STNR-Det is superior to previous methods. Extensive ablation studies confirm its effectiveness.

II. METHODOLOGY

A. Overview

In view of persistent limitations in bi-temporal feature interaction and the preservation of high-frequency transition-region details, we propose a Synergistic Temporal Nuance Reconstruction Detector (STNR-Det). It employs an end-to-end U-shaped encoder-decoder approach, which follows the sequence of feature extraction, synergistic comparison, and progressive reconstruction. As illustrated in Fig. 1, STNR-Det consists of a dual-branch Multi-Stage Siamese Encoder with shared weights enhanced by the SRCM Block [13], a Contrastive Synergistic Interaction (CSI) Block for bi-temporal feature interaction, and a Progressive Nuance Reconstitution (PNR) Decoder for detailed reconstruction.

B. Contrastive Synergistic Interaction Block

To address limited multi-scale and cross-temporal interaction in change detection, we propose the Contrastive Synergistic Interaction (CSI) block. As shown in Fig. 2, CSI first generates an initial change cue through bi-temporal differencing and refines it with the Contrastive Difference Aggregator (CDA) to emphasize change-salient regions. The enhanced

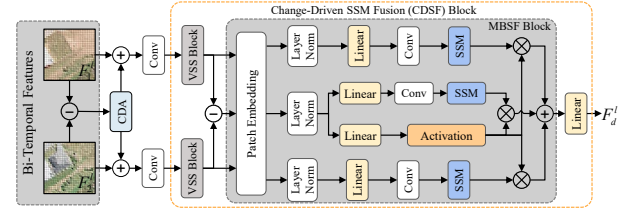


Fig. 2. Detailed architecture of the Contrastive Synergistic Interaction (CSI) block.

features are then fused and refined by the Change-Driven SSM Fusion (CDSF) and Multi-Branch SSM Fusion (MBSF) blocks. As a result, CSI improves feature discriminability, suppresses spurious changes, and yields sharper boundaries and more stable detection results.

CSI first computes the contrast cue by element-wise subtraction of the bi-temporal features F_1^l and F_2^l :

$$\mathbf{D}_0^l = \mathbf{F}_1^l \ominus \mathbf{F}_2^l, \quad (1)$$

This cue is then processed by the Contrastive Difference Aggregator (CDA), which computes attention weights within a local window and generates the change-enhanced term $\mathbf{C}^l$:

$$\mathbf{C}^l = \text{CDA}(\mathbf{D}_0^l). \quad (2)$$

The change-conditioned features are injected into the two temporal streams:

$$\tilde{\mathbf{F}}_1^l = \mathbf{F}_1^l \oplus \mathbf{C}^l, \quad (3)$$

$$\tilde{\mathbf{F}}_2^l = \mathbf{F}_2^l \oplus \mathbf{C}^l. \quad (4)$$

These features are passed through the Change-Driven SSM Fusion (CDSF) module and the Multi-Branch SSM Fusion (MBSF) module. The output of CDSF generates the stage-wise difference $\mathbf{D}^l$ by contrasting the outputs of the VSS blocks:

$$\mathbf{D}^l = \mathbf{G}_1^l \ominus \mathbf{G}_2^l, \quad (5)$$

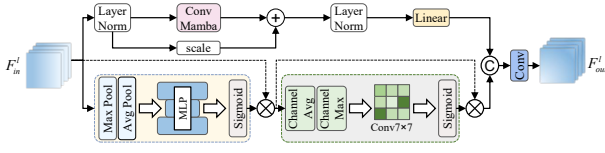


Fig. 3. Architecture of the PNR Block.

where $\mathbf{G}_1^l$ and $\mathbf{G}_2^l$ denote the outputs of the VSS blocks for two temporal streams. This difference is then embedded into a sequence of tokens:

$$\mathbf{T}^l = \text{PE}(\mathbf{D}^l). \quad (6)$$

In the MBSF module, the tokens are processed by three SSM branches, where each branch updates its state via the state-space equations:

$$\begin{bmatrix} \mathbf{h}_{t+1}^{(b)} \\ \mathbf{s}_t^{(b)} \end{bmatrix} = \begin{bmatrix} \mathbf{A}^{(b)} & \mathbf{B}^{(b)} \\ \mathbf{C}^{(b)} & \mathbf{D}^{(b)} \end{bmatrix} \begin{bmatrix} \mathbf{h}_t^{(b)} \\ \mathbf{x}_t^{(b)} \end{bmatrix}, \quad b \in \{D, F_1, F_2\}. \quad (7)$$

The outputs of the top, middle, and bottom SSM branches at the l -th stage are denoted by $\mathbf{U}_t^l$, $\mathbf{U}_m^l$ and $\mathbf{U}_b^l$, corresponding to the F_1 , D , and F_2 feature streams, respectively. The gating vector $\mathbf{g}^l$ modulates the response of the middle branch $\mathbf{U}_m^l$:

$$\mathbf{M}^l = \mathbf{U}_m^l \otimes \mathbf{g}^l. \quad (8)$$

Finally, the outputs from all three branches are aggregated to produce the final output:

$$\mathbf{Z}^l = (\mathbf{U}_t^l \otimes \mathbf{M}^l) \oplus \mathbf{M}^l \oplus (\mathbf{U}_b^l \otimes \mathbf{M}^l). \quad (9)$$

After a linear projection, the final stage output $\mathbf{F}_d^l$ is produced.

C. Progressive Nuance Reconstitution Decoder

To address insufficient detail reconstruction and boundary information loss, we propose the Progressive Nuance Reconstitution (PNR) Decoder. It progressively aggregates multi-level features to restore high-frequency details and improve boundary consistency. As shown in Fig. 3, the PNR Block combines channel and spatial attention with semantic refinement, enabling more accurate reconstruction of fine structures and change boundaries.

The PNR block takes the input feature $\mathbf{F}_{in}^l$ and processes it with two branches. In the lower branch, adaptive max pooling and adaptive average pooling are first applied to extract global channel descriptors:

$$\mathbf{s}_m = P_m(\mathbf{F}_{in}^l), \quad \mathbf{s}_a = P_a(\mathbf{F}_{in}^l), \quad (10)$$

where P_m and P_a denote adaptive max pooling and adaptive average pooling, respectively. These descriptors are then fed into a lightweight MLP for channel reduction and semantic reorganization. Based on the channel-refined feature $\mathbf{F}_c^l$, a spatial attention weight $\mathbf{W}_s$ is further generated to recalibrate the feature through elementwise multiplication:

$$\mathbf{Y}^l = \mathbf{W}_s \odot \mathbf{F}_c^l. \quad (11)$$

Meanwhile, the semantic refinement branch employs layer normalization, ConvMamba, and a linear projection to capture dynamic patterns, followed by residual addition and scaling:

$$\mathbf{F}_m = \alpha \cdot \text{Linear} \left(\text{LN} \left(\text{CM}(\text{LN}(\mathbf{F}_{in}^l)) + \mathbf{F}_{in}^l \right) \right). \quad (12)$$

Finally, the outputs of the two branches are concatenated and fused by a convolution layer to obtain the final output feature:

$$\mathbf{F}_{out}^l = \text{Conv}(\text{Concat}(\mathbf{Y}^l, \mathbf{F}_m)). \quad (13)$$

III. EXPERIMENT

A. Datasets and Implementation Setup

1) *Datasets*: Extensive experiments are conducted on three representative change detection benchmarks, including WHU-CD [20], LEVIR-CD [21] and SYSU-CD [22].

2) *Evaluation Metrics*: We evaluate the model using five standard metrics for binary change detection: precision (Pre.), recall (Rec.), F1-score (F1), intersection over union (IoU) and overall accuracy (OA).

3) *Implementation Details*: The method is implemented in PyTorch and trained on an NVIDIA RTX 5090D GPU. Adam is adopted with an initial learning rate of 1×10^{-4} , $\beta_1 = 0.9$, and $\beta_2 = 0.999$. The mini-batch size is 6, and the model is trained for 200 epochs. The training loss combines cross-entropy loss and Dice loss with equal weights.

B. Performance Comparison

To evaluate the effectiveness of STNR-Det, we compare it with representative state-of-the-art methods on the WHU-CD, LEVIR-CD, and SYSU-CD datasets, including the CNN-based methods FC-EF [6], FC-Siam-Diff [6], FC-Siam-Conc [6], SNU-Net [7] and S²ENet [14]; the Transformer-based methods BIT [8], PA-Former [15], ChangeFormer [9], FarmCD [16], and STRobustNet [17]; and the Mamba-based methods ChangeMamba-B [18], CTTF [19], and CD-Mamba [13]. Quantitative evaluation is conducted using five standard metrics, along with model complexity analysis in terms of Params and FLOPs, while qualitative comparison is performed through visual inspection of detection results.

As shown in Table I, all results are reported as percentages (%), while Params and FLOPs are reported in M and G, respectively. Optimal and sub-optimal outcomes are highlighted in bold and underlined, respectively. STNR-Det achieves the best F1 scores on all three datasets, reaching 95.63% on WHU-CD, 91.76% on LEVIR-CD, and 84.72% on SYSU-CD, demonstrating its superior performance and strong robustness in binary change detection. The visualization results in Fig. 4 further show that STNR-Det can more accurately capture subtle changes and handle complex scenes than competing methods such as FC-EF, BIT, and ChangeMamba-B. In particular, it produces clearer and more precise detection results in urban and vegetation regions. Overall, both quantitative and qualitative results verify the effectiveness of STNR-Det for binary change detection tasks.

TABLE I
QUANTITATIVE COMPARISONS IN TERMS OF **PARAMS** (M), **FLOPS** (G), **PRE.** (%), **REC.** (%), **F1** (%), **IoU** (%), AND **OA** (%) ON THREE PUBLIC REMOTE SENSING CHANGE DETECTION DATASETS. THE SYMBOL “-” INDICATES UNAVAILABLE RESULTS.

Methods	Params	FLOPs	WHU-CD					LEVIR-CD					SYSU-CD				
			Pre. / Rec. / F1 / IoU / OA	Pre. / Rec. / F1 / IoU / OA	Pre. / Rec. / F1 / IoU / OA	Pre. / Rec. / F1 / IoU / OA	Pre. / Rec. / F1 / IoU / OA										
<i>CNN-based Approaches</i>																	
FC-EF [6]	1.35	3.11	93.69	87.87	90.69	82.96	99.28	91.17	86.59	88.82	79.88	98.89	75.93	81.03	78.40	64.47	89.47
FC-Siam-Diff	1.35	4.25	90.17	90.05	90.11	82.00	99.22	91.57	87.64	89.57	81.10	98.96	81.19	76.43	78.73	64.93	90.26
FC-Siam-Conc	1.55	4.86	84.56	91.90	88.08	78.69	99.01	91.83	88.51	90.14	82.04	99.01	80.12	78.92	79.52	66.00	90.41
SNUNet [7]	1.35	4.25	83.34	90.88	86.95	76.92	98.92	89.93	87.45	88.67	79.65	98.86	88.44	61.63	72.64	57.04	89.05
S ² ENet [14]	17.59	24.92	95.67	93.09	94.36	89.33	-	<u>92.03</u>	<u>91.30</u>	<u>91.67</u>	<u>84.61</u>	-	86.07	81.58	82.08	<u>72.07</u>	-
<i>Transformer-based Approaches</i>																	
BIT [8]	11.94	8.76	91.58	92.55	92.06	85.29	99.37	91.55	88.54	90.02	81.84	99.00	79.04	75.96	77.47	63.23	89.58
PA-Former [15]	16.13	10.88	92.87	90.85	91.85	84.92	99.36	91.76	87.59	89.63	81.21	98.97	82.26	73.84	77.82	63.69	90.07
ChangeFormer [9]	41.03	138.77	91.85	84.11	87.81	78.28	99.07	90.65	87.20	88.89	80.00	98.89	80.04	77.06	78.52	64.64	90.06
FarmCD [16]	47.91	14.91	<u>95.89</u>	<u>93.34</u>	<u>94.60</u>	<u>89.75</u>	<u>99.58</u>	91.98	90.59	91.28	83.96	99.12	<u>87.64</u>	78.02	82.55	70.28	<u>92.22</u>
STRobustNet [17]	10.52	17.67	91.86	92.07	91.96	<u>85.12</u>	99.36	90.94	89.98	90.45	82.57	99.03	77.79	79.92	78.84	65.07	89.88
<i>Mamba-based Approaches</i>																	
ChangeMamba-B [18]	48.56	29.43	93.14	91.63	92.38	85.83	99.40	90.55	89.10	89.82	81.52	98.97	82.89	76.90	79.78	66.36	90.81
CTTF [19]	16.21	3.83	95.62	92.65	94.11	88.88	99.47	92.30	90.30	91.29	83.98	99.25	83.70	81.90	<u>82.79</u>	70.64	91.97
CDMamba [13]	6.84	49.68	95.58	92.01	93.76	88.26	99.51	91.43	90.08	90.75	83.07	99.06	81.92	<u>76.58</u>	<u>79.16</u>	65.51	90.49
Ours	5.67	25.41	96.34	94.93	95.63	91.62	99.66	91.35	92.17	91.76	84.77	<u>99.16</u>	84.98	84.45	84.72	73.49	92.81

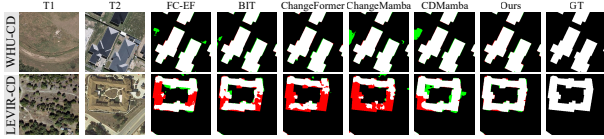


Fig. 4. Visualization of results produced by different methods on the test sets of WHU-CD and LEVIR-CD. White, Black, Red, and Green are true positive, true negative, false positive, and false negative regions, respectively.

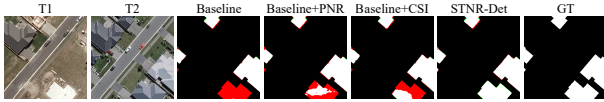


Fig. 5. Visualization of the ablation study on the WHU-CD test set. White, Black, Red, and Green are true positive, true negative, false positive, and false negative regions, respectively.

TABLE II
ABLATION STUDY ON DIFFERENT COMPONENTS (CSI AND PNR) ON WHU-CD DATASET.

Model	WHU-CD				
	Pre. (%)	Rec. (%)	F1 (%)	IoU (%)	OA (%)
Baseline	94.97	90.77	92.83	86.61	99.44
Baseline+PNR	95.74	93.10	94.40	89.40	99.56
Baseline+CSI	95.98	94.95	95.46	91.32	99.64
Baseline+CSI+PNR	96.34	94.93	95.63	91.62	99.66

C. Ablation Studies

To validate the effectiveness of each component of the proposed STNR-Det, a series of ablation experiments were conducted on the WHU-CD dataset.

1) *Effectiveness of Each Component*: The ablation results on the WHU-CD dataset are shown in Table II. The baseline adopts a simple architecture with Scaled Residual ConvMamba [13] as the core module in both the encoder and decoder. After introducing CSI and PNR, the performance improves consistently: CSI strengthens cross-temporal feature interaction, while PNR preserves details and refines boundaries. When combined, the two modules achieve the best

TABLE III
ABLATION STUDY ON DIFFERENT COMPONENTS OF CSI BLOCK WITHOUT PNR BLOCK ON WHU-CD DATASET.

Baseline	CDA	VSS	MBSF	WHU-CD				
				Pre. (%)	Rec. (%)	F1 (%)	IoU (%)	OA (%)
✓	×	×	×	94.97	90.77	92.83	86.61	99.44
✓	✓	×	×	95.34	92.91	94.11	88.88	99.54
✓	×	×	✓	96.87	92.84	94.81	90.14	99.60
✓	×	✓	✓	96.52	93.33	94.90	90.29	99.60
✓	✓	✓	✓	95.98	94.95	95.46	91.32	99.64

TABLE IV
ABLATION STUDY ON DIFFERENT INTERACTION MODULES OF CSI BLOCK.

Model	WHU-CD				
	Pre. (%)	Rec. (%)	F1 (%)	IoU (%)	OA (%)
BiFPN [23]	91.56	86.43	88.93	80.06	99.16
IHP [24]	92.57	87.65	90.04	81.89	99.24
FreqFusion [25]	92.10	87.74	89.87	81.60	99.22
Mamba [10]	86.54	85.51	86.02	75.47	98.91
AGLGF [13]	93.33	89.40	91.32	84.03	99.33
CSI	96.34	94.93	95.63	91.62	99.66

overall performance, indicating their effectiveness and complementarity. The visual results in Fig. 5 further confirm this conclusion. Compared with the baseline, models with CSI or PNR produce fewer false positives and false negatives, while STNR-Det generates prediction maps closest to the ground truth, with cleaner regions and more accurate boundaries.

2) *Effectiveness of CSI Components*: To better understand the performance gain brought by CSI, we further ablate its submodules, with results shown in Table III. The results indicate that MBSF, VSS, and CDA all contribute to performance improvement, while combinations including MBSF perform better overall. The full combination achieves the best results, demonstrating the complementarity of the three submodules in enhancing bi-temporal feature fusion.

3) *Effectiveness of CSI Block*: On WHU-CD, we validate CSI through a replacement study, with results shown in Table IV. Compared with BiFPN [23], IHP [24], FreqFusion [25], Mamba [10], and AGLGF [13], CSI achieves

TABLE V
ABLATION STUDY ON DIFFERENT STAGES OF CDSF BLOCK ON WHU-CD DATASET.

Stage1	Stage2	Stage3	Stage4	WHU-CD				
				Pre. (%)	Rec. (%)	F1 (%)	IoU (%)	OA (%)
×	×	×	×	94.97	90.77	92.83	86.61	99.44
✓	×	×	×	96.09	93.96	95.01	90.50	99.61
✓	✓	×	×	96.34	94.93	95.63	91.62	99.66
✓	✓	✓	×	96.51	93.27	94.86	90.22	99.60
✓	✓	✓	✓	96.07	93.02	94.52	89.62	99.57

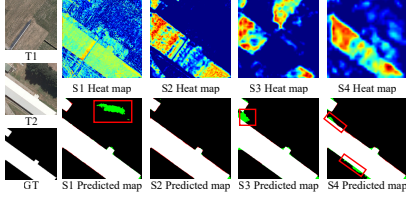


Fig. 6. Visualization results of the different stages of CDSF Block on the WHU-CD test set. Red denotes higher attention values and blue denotes lower values in heat maps. In predicted maps, White, Black, Red, and Green are true positive, true negative, false positive, and false negative regions, respectively.

the highest F1 and IoU, demonstrating its superiority in bi-temporal feature fusion.

4) *Effectiveness of Different Stages of CDSF Block*: The ablation results are reported in Table V. Incorporating CDSF improves the model performance, and the best results are obtained when CDSF Blocks are placed in the first two stages. As shown in Fig. 6, this setting produces more accurate responses in change regions and yields fewer false positives and missed detections. In contrast, introducing CDSF into deeper stages brings no further improvement and may weaken fine details and boundary quality. Therefore, CDSF Blocks are deployed in stages 1 and 2 in the final model.

IV. CONCLUSION

This work presents STNR-Det to address insufficient cross-temporal interaction and transition-region detail loss in RSCD. CSI enhances bi-temporal and multi-scale feature alignment through contrast-guided interaction, improving true-change discrimination and suppressing pseudo changes. PNR progressively restores decoder-side details, leading to sharper boundaries and more complete change regions. Experiments on three benchmarks demonstrate that STNR-Det outperforms representative CNN-, Transformer-, and Mamba-based methods, while ablation studies verify the complementary roles of CSI and PNR. Future work will explore its extension to multi-class, multi-modal, and multi-temporal change detection.

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